

VESSEL ANALYSIS REPORT

vaf(%)	branch points count	tip point count	vein count	total vein length	average vein length
2.725	141.0	126.0	276.0	7974.955	28.895

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
1	[240,7]	[210,9]	29.828	BRANCH	BRANCH
2	[239,47]	[240,7]	41.071	TIP	BRANCH
3	[250,0]	[240,7]	11.485	TIP	BRANCH
4	[210,9]	[131,80]	129.669	BRANCH	BRANCH
5	[210,9]	[149,12]	69.941	BRANCH	BRANCH
6	[149,12]	[84,54]	105.255	BRANCH	BRANCH
7	[162,0]	[149,12]	16.556	TIP	BRANCH
8	[444,14]	[472,18]	28.657	BRANCH	BRANCH
9	[428,1]	[444,14]	19.971	TIP	BRANCH
10	[444,14]	[421,52]	47.87	BRANCH	BRANCH
11	[358,0]	[350,17]	20.485	TIP	BRANCH
12	[305,31]	[350,17]	49.799	TIP	BRANCH
13	[350,17]	[387,51]	58.456	BRANCH	BRANCH
14	[472,18]	[500,50]	51.385	BRANCH	BRANCH
15	[472,18]	[474,66]	57.113	BRANCH	BRANCH
16	[71,6]	[62,21]	19.485	TIP	BRANCH
17	[62,21]	[12,55]	105.87	BRANCH	BRANCH
18	[62,21]	[63,40]	18.414	BRANCH	BRANCH
19	[63,40]	[84,54]	25.799	BRANCH	BRANCH
20	[63,40]	[38,61]	32.698	BRANCH	BRANCH
21	[285,36]	[286,42]	5.414	TIP	BRANCH
22	[324,54]	[286,42]	41.971	TIP	BRANCH
23	[280,51]	[286,42]	10.657	TIP	BRANCH
24	[509,58]	[500,50]	11.314	TIP	BRANCH
25	[500,50]	[474,66]	31.627	BRANCH	BRANCH
26	[387,51]	[399,55]	12.657	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
27	[387,51]	[342,72]	52.698	BRANCH	BRANCH
28	[421,52]	[442,77]	34.456	BRANCH	BRANCH
29	[421,52]	[399,55]	22.243	BRANCH	BRANCH
30	[84,54]	[84,60]	5.0	BRANCH	BRANCH
31	[12,55]	[38,61]	27.485	BRANCH	BRANCH
32	[12,55]	[19,116]	73.698	BRANCH	BRANCH
33	[399,55]	[395,72]	17.657	BRANCH	BRANCH
34	[84,60]	[110,76]	32.213	BRANCH	BRANCH
35	[84,60]	[52,90]	45.77	BRANCH	BRANCH
36	[38,61]	[50,90]	32.971	BRANCH	BRANCH
37	[474,66]	[471,77]	11.243	BRANCH	BRANCH
38	[342,72]	[348,126]	72.113	BRANCH	BRANCH
39	[330,59]	[342,72]	17.142	TIP	BRANCH
40	[381,72]	[395,72]	13.828	TIP	BRANCH
41	[395,72]	[399,81]	9.243	BRANCH	BRANCH
42	[110,76]	[131,80]	21.657	BRANCH	BRANCH
43	[108,82]	[110,76]	5.828	TIP	BRANCH
44	[442,77]	[444,77]	1.0	BRANCH	BRANCH
45	[442,77]	[399,81]	45.728	BRANCH	BRANCH
46	[444,77]	[471,77]	26.0	BRANCH	BRANCH
47	[444,77]	[454,148]	104.083	BRANCH	BRANCH
48	[471,77]	[489,90]	21.971	BRANCH	BRANCH
49	[131,80]	[146,152]	80.527	BRANCH	BRANCH
50	[399,81]	[381,103]	30.971	BRANCH	BRANCH
51	[50,90]	[52,90]	1.0	BRANCH	BRANCH
52	[50,90]	[48,95]	4.414	BRANCH	BRANCH
53	[52,90]	[80,133]	58.284	BRANCH	BRANCH
54	[506,78]	[489,90]	20.971	TIP	BRANCH
55	[489,90]	[474,149]	67.627	BRANCH	BRANCH
56	[48,95]	[66,142]	54.042	BRANCH	BRANCH
57	[48,95]	[19,116]	36.698	BRANCH	BRANCH
58	[381,103]	[348,126]	42.698	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
59	[388,107]	[381,103]	7.243	TIP	BRANCH
60	[19,116]	[29,134]	21.142	BRANCH	BRANCH
61	[347,135]	[348,126]	8.414	TIP	BRANCH
62	[231,125]	[270,129]	41.899	TIP	BRANCH
63	[279,127]	[270,129]	8.828	TIP	BRANCH
64	[236,164]	[270,129]	51.598	TIP	BRANCH
65	[87,131]	[80,133]	6.828	TIP	BRANCH
66	[80,133]	[66,142]	17.314	BRANCH	BRANCH
67	[29,134]	[60,147]	35.385	BRANCH	BRANCH
68	[29,134]	[30,142]	7.414	BRANCH	BRANCH
69	[30,142]	[31,144]	2.828	BRANCH	PATH
70	[66,142]	[60,147]	8.243	BRANCH	BRANCH
71	[60,147]	[53,164]	18.899	BRANCH	BRANCH
72	[413,147]	[424,139]	13.314	BRANCH	PATH
73	[413,147]	[412,162]	14.414	BRANCH	BRANCH
74	[413,147]	[339,155]	82.527	BRANCH	BRANCH
75	[454,148]	[474,149]	19.414	BRANCH	BRANCH
76	[454,148]	[426,176]	39.355	BRANCH	BRANCH
77	[474,149]	[474,170]	20.0	BRANCH	BRANCH
78	[165,142]	[157,151]	11.314	TIP	BRANCH
79	[157,151]	[146,152]	10.414	BRANCH	BRANCH
80	[171,160]	[157,151]	17.314	TIP	BRANCH
81	[146,152]	[136,159]	11.485	BRANCH	BRANCH
82	[339,155]	[339,179]	24.657	BRANCH	BRANCH
83	[328,153]	[339,155]	10.414	TIP	BRANCH
84	[131,180]	[136,159]	22.899	TIP	BRANCH
85	[119,155]	[136,159]	18.071	TIP	BRANCH
86	[324,154]	[318,161]	8.485	TIP	BRANCH
87	[309,159]	[318,161]	8.828	TIP	BRANCH
88	[320,164]	[318,161]	2.828	TIP	BRANCH
89	[412,162]	[387,179]	32.799	BRANCH	BRANCH
90	[412,162]	[418,175]	14.071	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
91	[53,164]	[34,273]	149.811	BRANCH	BRANCH
92	[53,164]	[38,274]	146.711	BRANCH	BRANCH
93	[474,170]	[435,187]	45.042	BRANCH	BRANCH
94	[474,170]	[493,186]	27.142	BRANCH	BRANCH
95	[418,175]	[426,176]	7.414	BRANCH	BRANCH
96	[418,175]	[398,186]	23.142	BRANCH	BRANCH
97	[426,176]	[435,187]	14.899	BRANCH	BRANCH
98	[339,179]	[369,194]	35.213	BRANCH	BRANCH
99	[339,179]	[330,215]	38.314	BRANCH	BRANCH
100	[387,179]	[398,186]	12.899	BRANCH	BRANCH
101	[387,179]	[369,194]	24.385	BRANCH	BRANCH
102	[398,186]	[402,217]	31.657	BRANCH	BRANCH
103	[508,174]	[493,186]	18.971	TIP	BRANCH
104	[493,186]	[497,216]	30.657	BRANCH	BRANCH
105	[435,187]	[407,217]	42.941	BRANCH	BRANCH
106	[369,194]	[369,205]	10.0	BRANCH	BRANCH
107	[369,205]	[330,215]	43.799	BRANCH	BRANCH
108	[369,205]	[387,229]	30.042	BRANCH	BRANCH
109	[330,215]	[320,219]	10.657	BRANCH	BRANCH
110	[497,216]	[464,226]	36.142	BRANCH	BRANCH
111	[511,232]	[497,216]	20.385	TIP	BRANCH
112	[402,217]	[407,217]	4.0	BRANCH	BRANCH
113	[402,217]	[387,229]	18.556	BRANCH	BRANCH
114	[407,217]	[423,240]	29.213	BRANCH	BRANCH
115	[259,225]	[320,219]	69.941	TIP	BRANCH
116	[320,219]	[304,252]	38.627	BRANCH	BRANCH
117	[464,226]	[429,240]	39.799	BRANCH	BRANCH
118	[464,226]	[465,245]	18.414	BRANCH	BRANCH
119	[387,229]	[386,243]	13.414	BRANCH	BRANCH
120	[423,240]	[429,240]	5.0	BRANCH	BRANCH
121	[423,240]	[400,253]	26.971	BRANCH	BRANCH
122	[429,240]	[456,262]	39.213	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
123	[386,243]	[388,250]	7.414	BRANCH	BRANCH
124	[365,252]	[386,243]	23.728	TIP	BRANCH
125	[465,245]	[456,262]	20.314	BRANCH	BRANCH
126	[465,245]	[498,286]	82.284	BRANCH	BRANCH
127	[388,250]	[400,253]	12.243	BRANCH	BRANCH
128	[388,250]	[352,286]	56.941	BRANCH	BRANCH
129	[279,251]	[304,252]	25.243	TIP	BRANCH
130	[304,252]	[292,285]	36.971	BRANCH	BRANCH
131	[400,253]	[411,286]	36.556	BRANCH	BRANCH
132	[456,262]	[456,274]	11.0	BRANCH	BRANCH
133	[34,273]	[38,274]	3.414	BRANCH	BRANCH
134	[34,273]	[17,280]	18.485	BRANCH	BRANCH
135	[38,274]	[39,295]	20.414	BRANCH	BRANCH
136	[456,274]	[465,284]	13.899	BRANCH	BRANCH
137	[456,274]	[411,286]	49.799	BRANCH	BRANCH
138	[228,244]	[235,277]	35.728	TIP	BRANCH
139	[214,276]	[235,277]	20.414	TIP	BRANCH
140	[244,285]	[235,277]	11.899	TIP	BRANCH
141	[0,317]	[17,280]	46.799	TIP	BRANCH
142	[1,264]	[17,280]	23.556	TIP	BRANCH
143	[465,284]	[498,286]	32.828	BRANCH	BRANCH
144	[465,284]	[462,293]	8.828	BRANCH	BRANCH
145	[292,285]	[272,291]	21.485	BRANCH	BRANCH
146	[292,285]	[320,312]	44.213	BRANCH	BRANCH
147	[352,286]	[339,312]	31.556	BRANCH	BRANCH
148	[325,255]	[352,286]	52.627	TIP	BRANCH
149	[411,286]	[400,322]	39.556	BRANCH	BRANCH
150	[498,286]	[500,300]	13.828	BRANCH	BRANCH
151	[269,290]	[272,291]	2.414	TIP	BRANCH
152	[270,305]	[272,291]	13.828	TIP	BRANCH
153	[472,308]	[462,293]	18.728	TIP	BRANCH
154	[462,293]	[438,326]	43.113	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
155	[62,304]	[39,295]	25.728	TIP	BRANCH
156	[35,303]	[39,295]	8.243	TIP	BRANCH
157	[500,300]	[487,323]	27.971	BRANCH	BRANCH
158	[510,313]	[500,300]	15.728	TIP	BRANCH
159	[80,292]	[95,302]	28.142	TIP	BRANCH
160	[102,305]	[95,302]	7.243	TIP	BRANCH
161	[95,302]	[39,350]	78.397	BRANCH	BRANCH
162	[320,312]	[339,312]	18.0	BRANCH	BRANCH
163	[310,320]	[320,312]	14.071	TIP	BRANCH
164	[339,312]	[343,340]	28.657	BRANCH	BRANCH
165	[400,322]	[371,344]	40.042	BRANCH	BRANCH
166	[400,322]	[415,337]	20.385	BRANCH	BRANCH
167	[487,323]	[457,329]	31.485	BRANCH	BRANCH
168	[487,323]	[497,345]	25.314	BRANCH	BRANCH
169	[438,326]	[423,337]	19.142	BRANCH	BRANCH
170	[438,326]	[457,329]	18.828	BRANCH	BRANCH
171	[457,329]	[442,351]	32.728	BRANCH	BRANCH
172	[415,337]	[423,337]	7.0	BRANCH	BRANCH
173	[415,337]	[371,353]	51.456	BRANCH	BRANCH
174	[423,337]	[442,351]	24.971	BRANCH	BRANCH
175	[343,340]	[371,344]	28.657	BRANCH	BRANCH
176	[343,340]	[334,355]	17.314	BRANCH	BRANCH
177	[371,344]	[371,353]	8.0	BRANCH	BRANCH
178	[497,345]	[499,348]	3.414	BRANCH	BRANCH
179	[497,345]	[452,368]	54.113	BRANCH	BRANCH
180	[510,366]	[499,348]	22.142	TIP	BRANCH
181	[499,348]	[500,408]	72.184	BRANCH	BRANCH
182	[4,362]	[39,350]	40.728	TIP	BRANCH
183	[69,402]	[39,350]	75.426	TIP	BRANCH
184	[297,319]	[249,350]	65.941	TIP	BRANCH
185	[249,350]	[277,434]	143.983	BRANCH	BRANCH
186	[249,350]	[290,363]	45.556	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
187	[442,351]	[449,368]	18.485	BRANCH	BRANCH
188	[371,353]	[368,369]	15.828	BRANCH	BRANCH
189	[334,355]	[290,363]	50.456	BRANCH	BRANCH
190	[334,355]	[338,371]	16.657	BRANCH	BRANCH
191	[290,363]	[289,386]	22.414	BRANCH	BRANCH
192	[449,368]	[452,368]	2.0	BRANCH	BRANCH
193	[449,368]	[418,386]	37.042	BRANCH	BRANCH
194	[452,368]	[469,409]	47.042	BRANCH	BRANCH
195	[368,369]	[359,375]	10.071	BRANCH	BRANCH
196	[379,377]	[368,369]	12.899	TIP	BRANCH
197	[338,371]	[359,375]	21.657	BRANCH	BRANCH
198	[325,384]	[338,371]	18.556	TIP	BRANCH
199	[359,375]	[359,390]	14.0	BRANCH	BRANCH
200	[289,386]	[277,434]	59.213	BRANCH	BRANCH
201	[289,386]	[312,392]	24.071	BRANCH	BRANCH
202	[418,386]	[359,390]	63.799	BRANCH	BRANCH
203	[418,386]	[427,406]	22.899	BRANCH	BRANCH
204	[359,390]	[340,408]	27.799	BRANCH	BRANCH
205	[312,392]	[317,404]	13.071	BRANCH	BRANCH
206	[318,387]	[312,392]	6.657	TIP	BRANCH
207	[317,404]	[340,408]	23.657	BRANCH	BRANCH
208	[315,410]	[317,404]	5.414	TIP	BRANCH
209	[435,399]	[427,406]	9.899	TIP	BRANCH
210	[427,406]	[426,410]	3.414	BRANCH	BRANCH
211	[340,408]	[340,429]	20.0	BRANCH	BRANCH
212	[500,408]	[486,415]	15.899	BRANCH	BRANCH
213	[511,419]	[500,408]	14.142	TIP	BRANCH
214	[469,409]	[479,415]	11.485	BRANCH	BRANCH
215	[469,409]	[458,424]	18.142	BRANCH	BRANCH
216	[426,410]	[418,415]	9.071	BRANCH	BRANCH
217	[426,410]	[443,424]	22.556	BRANCH	BRANCH
218	[418,415]	[391,427]	30.971	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
219	[418,415]	[419,442]	27.657	BRANCH	BRANCH
220	[479,415]	[486,415]	6.0	BRANCH	BRANCH
221	[479,415]	[469,430]	17.728	BRANCH	BRANCH
222	[486,415]	[494,445]	32.314	BRANCH	BRANCH
223	[443,424]	[458,424]	14.0	BRANCH	BRANCH
224	[439,429]	[443,424]	5.657	TIP	BRANCH
225	[458,424]	[469,430]	13.071	BRANCH	BRANCH
226	[391,427]	[345,430]	47.071	BRANCH	BRANCH
227	[391,427]	[391,437]	9.0	BRANCH	BRANCH
228	[340,429]	[345,430]	4.414	BRANCH	BRANCH
229	[340,429]	[312,433]	28.243	BRANCH	BRANCH
230	[345,430]	[385,452]	83.87	BRANCH	BRANCH
231	[469,430]	[480,446]	19.556	BRANCH	BRANCH
232	[312,420]	[312,433]	12.0	TIP	BRANCH
233	[312,433]	[277,434]	34.414	BRANCH	BRANCH
234	[391,437]	[419,442]	29.071	BRANCH	BRANCH
235	[391,437]	[385,452]	16.071	BRANCH	BRANCH
236	[419,442]	[436,468]	31.627	BRANCH	BRANCH
237	[511,456]	[494,445]	21.728	TIP	BRANCH
238	[494,445]	[480,446]	13.414	BRANCH	BRANCH
239	[480,446]	[440,468]	49.87	BRANCH	BRANCH
240	[385,452]	[386,477]	24.414	BRANCH	BRANCH
241	[436,468]	[440,468]	3.0	BRANCH	BRANCH
242	[436,468]	[386,477]	54.799	BRANCH	BRANCH
243	[447,477]	[440,468]	10.485	TIP	BRANCH
244	[376,485]	[386,477]	12.899	TIP	BRANCH
245	[406,14]	[414,20]	10.485	TIP	TIP
246	[249,64]	[258,83]	22.728	TIP	TIP
247	[262,86]	[299,88]	41.971	TIP	TIP
248	[211,103]	[225,108]	16.071	TIP	TIP
249	[96,112]	[94,122]	10.828	TIP	TIP
250	[409,123]	[424,136]	20.385	TIP	PATH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
251	[104,160]	[106,239]	93.426	TIP	TIP
252	[281,167]	[277,171]	5.657	TIP	TIP
253	[170,207]	[116,291]	115.154	TIP	TIP
254	[0,212]	[2,221]	10.657	TIP	TIP
255	[209,237]	[210,239]	2.414	TIP	TIP
256	[141,241]	[124,242]	18.243	TIP	TIP
257	[199,267]	[194,277]	12.071	TIP	TIP
258	[181,316]	[137,341]	56.355	TIP	TIP
259	[213,319]	[220,321]	8.657	TIP	TIP
260	[217,341]	[216,343]	2.414	TIP	TIP
261	[130,348]	[105,352]	27.485	TIP	TIP
262	[26,378]	[21,382]	6.657	TIP	TIP
263	[102,387]	[107,389]	5.828	TIP	TIP
264	[55,393]	[59,397]	6.243	TIP	TIP
265	[135,415]	[139,421]	7.657	TIP	TIP
266	[50,419]	[50,429]	11.657	TIP	TIP
267	[15,420]	[24,438]	22.314	TIP	TIP
268	[45,438]	[46,458]	22.071	TIP	TIP
269	[15,456]	[11,507]	58.799	TIP	TIP
270	[270,467]	[269,468]	1.414	TIP	TIP
271	[234,474]	[225,482]	12.314	TIP	TIP
272	[504,480]	[460,508]	60.284	TIP	TIP
273	[455,484]	[461,486]	6.828	TIP	TIP
274	[356,485]	[365,508]	27.314	TIP	TIP
275	[47,486]	[92,487]	51.213	TIP	TIP
276	[398,505]	[434,509]	39.899	TIP	TIP